ModelIt!

Teacher's Guide

How Do We Know the Earth Is 4.5 Billion Years Old?

Modeling Radioactive Decay, Radiometric Dating, and Deep Time

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-ESS1-5, HS-ESS1-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-ESS1-5, HS-ESS1-6

Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth's formation and early history.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze isotope ratio data to calculate rock ages and evaluate the accuracy of age determinations from multiple independent methods.
Disciplinary Core Idea	**ESS1.C The History of Planet Earth / PS1.C Nuclear Processes** — Radioactive decay provides a way to establish absolute ages for Earth's rocks and history. Nuclear decay rates are constant and provide reliable chronometers for geological time.
Crosscutting Concept	**Patterns** — Students identify the mathematical pattern of exponential decay in radioactive isotopes and use this pattern to calculate ages and predict isotope ratios at different points in geological time.

CAST Testing Alignment

MULTIPLE CHOICE:

A rock sample contains 25% of its original uranium-238 and 75% daughter lead-206. If the half-life of U-238 is 4.47 billion years, what is the age of this rock?

CONSTRUCTED RESPONSE:

Using your model, explain how scientists determined Earth's age is 4.567 billion years. Include the roles of radioactive decay, half-life, parent-daughter ratios, and multiple independent lines of evidence in your explanation. Why is it significant that meteorites, Moon rocks, and Earth's oldest minerals all yield consistent ages?

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The question 'How old is the Earth?' was unanswerable for most of human history. In 1862, Lord Kelvin calculated the Earth was 20-100 million years old based on cooling rates — but he didn't know about radioactivity. In 1953, Clair Patterson used uranium-lead dating of meteorites to determine the age of the Solar System — and Earth — at 4.55 billion years (later refined to 4.567 billion years). The key to this discovery was understanding that radioactive atoms are perfect clocks.

Radioactive Decay as a Clock

Radioactive decay is a quantum mechanical process where an unstable nucleus spontaneously transforms into a more stable configuration. This process follows a strict exponential decay curve: $N(t) = N_0 \times (1/2)^{(t/t½)}$. The half-life is absolutely constant — it cannot be changed by any known physical or chemical process. This constancy is what makes radioactive isotopes perfect clocks: they tick at the same rate regardless of temperature, pressure, or environment.

Uranium-Lead Dating

The gold standard for dating ancient rocks. Uranium-238 decays to lead-206 with a half-life of 4.47 billion years, while uranium-235 decays to lead-207 with a half-life of 704 million years. Having two independent decay chains in the same element provides a built-in cross-check (concordia). When both U-Pb systems give the same age, the result is considered highly reliable. This system is ideal for dating rocks from millions to billions of years old.

Multiple Lines of Evidence

Earth's age isn't based on a single measurement. Uranium-lead dating of meteorites gives 4.567 billion years. Potassium-argon dating of Moon rocks gives 4.51 billion years. Rubidium-strontium dating of early Earth rocks gives ages up to 4.0 billion years (the oldest Earth rocks were recycled, so none preserve the original formation age). Samarium-neodymium dating of Hadean zircons gives ages up to 4.4 billion years. All methods converge on the same answer.

The Scale of Deep Time

4.5 billion years is almost impossible to comprehend. If Earth's history were compressed to 24 hours, the first life appears at 4:00 AM, complex animals appear at 9:00 PM, dinosaurs dominate from 10:40 to 11:39 PM, and all of human history occurs in the last 0.2 seconds before midnight. Understanding deep time is essential for understanding geology, evolution, and the processes that shaped our planet.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify radioactive isotope amount, decay rate, rock layer age, geological event timing, and age estimate accuracy as the key components of the radiometric dating system.

E **Establish Relationships**

Students determine that decay rate is fixed by nuclear physics, the ratio of parent to daughter isotopes reveals how many half-lives have passed, and age is calculated by multiplying half-lives elapsed by the half-life duration.

V **Visualize & Model**

Students build a computational model showing exponential decay curves and using parent-daughter ratios to calculate rock ages across geological timescales.

E **Evaluate Outcomes**

Students run young rock, ancient rock, and multiple methods scenarios to calculate ages and discover that independent methods converge on consistent results.

R **Revise & Extend**

Students add isochron analysis, concordance testing, or metamorphic resetting to model more sophisticated dating techniques and their quality controls.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"4,500,000,000 years. That's the age of the Earth. But how can we possibly know that when the oldest human civilizations are only 6,000 years old?"*

Watch For

Write 4.5 billion on the board. Ask: How would you even BEGIN to measure something this old? What kind of clock runs for billions of years?

Slide 2: Learning Objectives

3 min

Say: *"Today you're going to learn about nature's most perfect clock — one that has been running since Earth was born, and we can read it."*

Watch For

Have students read objectives. Pre-teach 'half-life' and 'radiometric dating' with the coin-flip analogy.

Slide 3: Big Question

3 min

Say: *"No one was there 4.5 billion years ago. No written records. No photographs. So how can scientists be so CONFIDENT about this number?"*

Watch For

Students list what evidence they think exists. Most will underestimate the strength of the evidence.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of a clock that ticks for billions of years — and learning to read it."*

Watch For

Preview LEVER steps. Emphasize that this model is about EVIDENCE and CONVERGENCE of multiple independent measurements.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine how we read the atomic clock inside a rock?"*

Watch For

Guide sorting. Decay rate is the key external constant — set by physics, unchangeable by any known force.

Slide 6: Activity 2: Connections

8 min

Say: *"If you start with 100 uranium atoms and find 25 remaining and 75 lead atoms, how many half-lives passed?"*

Watch For

Work through the math: 100→50→25 = 2 half-lives × 4.47 billion years = 8.94 billion year old rock. Build the calculation chain.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's date three different rocks — and then check our answer with a completely different isotope system."*

Watch For

Students calculate ages from ratios. Then apply a second dating method to the same sample. Results should converge.

Slide 8: Discoveries

5 min

Say: *"Meteorites, Moon rocks, and Earth's oldest minerals all point to the same age. That's not a coincidence — that's evidence."*

Watch For

Show the convergence of multiple independent methods. Discuss why convergence is the strongest form of scientific evidence.

Slide 9: STEM Challenge

12 min

Say: *"A geological survey found rocks that might date a mass extinction. Design the lab protocol to nail down the age."*

Watch For

Groups design dating protocols specifying isotope systems, sample prep, quality controls, and cross-validation.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Decay Rate
- **INTERNAL:** Radioactive Isotope Amount, Rock Layer Age, Geological Event Timing, Age Estimate Accuracy

Relationships

- **Decay Rate to Radioactive Isotope Amount:** NEGATIVE (-) over time — The constant decay rate continuously reduces the amount of parent radioactive isotope in the rock. After each half-life period, exactly half of the remaining parent atoms have transformed into daughter atoms. This exponential decrease is the fundamental clock mechanism.
- **Radioactive Isotope Amount to Rock Layer Age:** INVERSE CALCULATION — The proportion of parent isotope remaining compared to the original amount directly reveals how many half-lives have elapsed. More parent remaining means younger rock; less parent remaining means older rock. The relationship follows: Age = half-life × $\log_2(N_0/N)$.
- **Rock Layer Age to Geological Event Timing:** DETERMINES — Dating specific rock layers places geological events — volcanic eruptions, asteroid impacts, mass extinctions, mountain building — on an absolute timeline. The age of a rock layer constrains the age of any fossils or geological features within or adjacent to it.

Key Discoveries

- Radioactive decay follows a precise mathematical curve — after one half-life, 50% remains; after two, 25%; after three, 12.5% — and this rate is constant regardless of environmental conditions
- The parent-to-daughter ratio is a built-in clock that starts ticking when a rock crystallizes, and by measuring this ratio today, we can calculate exactly how long the clock has been running
- Multiple independent dating methods (uranium-lead, potassium-argon, rubidium-strontium) applied to the same rocks produce the same ages — this cross-validation is powerful evidence that the methods work
- The oldest Earth rocks are about 4.0 billion years old, but meteorites (from the same solar system formation event) date to 4.567 billion years, and our Moon's oldest samples are 4.51 billion years — all pointing to the same age

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Carbon dating proves the Earth is billions of years old"	Carbon-14 dating is only useful for organic materials up to about 50,000 years old — it says nothing about Earth's age. Earth's age comes from uranium-lead, potassium-argon, and rubidium-strontium dating of rocks and meteorites, which work on billion-year timescales. Different isotope systems have different useful ranges: C-14 for recent biology, K-Ar for millions of years, U-Pb for billions of years.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Radioactive decay rates could have been different in the past"	Radioactive decay rates are determined by the fundamental constants of nuclear physics — the strong force and electromagnetic force. These constants have been measured to be unchanged across the observable universe (through spectral analysis of distant stars and galaxies) and across billions of years of cosmic history. To change decay rates, you would need to change the fundamental forces that hold atomic nuclei together — and there is zero evidence this has ever happened.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Radiometric dating is just one guess"	Earth's age is not based on one measurement but on THOUSANDS of independent measurements using DOZENS of different isotope systems on rocks from EVERY continent, plus meteorites and Moon samples. Multiple independent methods all converge on the same age: 4.567 billion years. The probability of this convergence being coincidental is effectively zero. This is one of the most thoroughly verified measurements in all of science.